Pixie

Pixie (https://px.dev/) is a CNCF project focusing on Kubernetes clusters, which we covered in chapter 6. It's not just about continuous profiling; Pixie also offers full-body requests (e.g., a JSON blob return value from an HTTP API) and resource and network metrics, among other useful items.

Because Pixie collects so much rich data, the telemetry data in Pixie is kept in memory and local to the Kubernetes cluster. To put it succinctly, you can think of Pixie as the "What is going on in my Kubernetes cluster right now?" tool. In terms of resource usage, Pixie claims to use less than 5% of cluster CPU at the maximum. Pixie can help you see your application profiles in a specific pod, across a node, and more, all without requiring you to make any code changes or redeploy your app. At the time of writing, Pixie supports continuous profiling for compiled languages such as Go, Rust, and Java.

Pyroscope

Pyroscope (https://pyroscope.io/) is an open source continuous profiling tool that comes with a rich client-side collection mechanism for several programming languages, including Go, Python, Java, Ruby, Rust, PHP, .NET, and Node.js, as well as eBPF support. In June 2023, Pyroscope was acquired by Grafana Labs (http://mng.bz/wvNP) and is, going forward, the foundation of the continuous profiling offering in Grafana Cloud.

The makers of Pyroscope recommend a hybrid approach for profile collection: that is, to use eBPF to profile the system (e.g., the VM or Kubernetes node) in combination with the specific language instrumentation per service. Additionally, Pyroscope sports FlameQL, a query language that enables you to explore profiling data. You can use the zero-install online environment at https://demo.pyroscope.io/ to try out Pyroscope or install it locally if you like.

Figure 9.5 shows an overview (and an overlay with a zoom into the baseline flame graph) of the Pyroscope UI. In the upper part of the figure, the time-based visualization of the profiles is displayed, and when you click any of the profiles, you can generate and zoom in on specific profiles (in this case, memory usage).